sEMG-Based End-to-End Continues Prediction of Human Knee Joint Angles Using the Tightly Coupled Convolutional Transformer Model

Tuanjie Liang, Ning Sun, Qiong Wang, Jingyu Bu, Long Li, Yuhao Chen, Menglin Cao, Jin Ma, and Tian Liu

***Abstract*—Wearable exoskeleton robots can promote the rehabilitation of patients with physical dysfunction. And improving human-computer interaction performance is a significant challenge for exoskeleton robots. The traditional feature extraction process based on surface Electromyography(sEMG) is complex and requires manual intervention, making real-time performance difficult to guarantee. In this study, we propose an end-to-end method to predict human knee joint angles based on sEMG signals using a tightly coupled convolutional transformer (TCCT) model. We first collected sEMG signals from 5 healthy subjects. Then, the envelope was extracted from the noise-removed sEMG signal and used as the input to the model. Finally, we developed the TCCT model to predict the knee joint angle after 100 ms. For the prediction performance, we used the Root Mean Square Error(RMSE), Pearson Correlation Coefficient(CC), and Adjustment R^2 as metrics to evaluate the error between the actual knee angle and the predicted knee angle. The results show that the model can predict the human knee angle quickly and accurately. The mean RMSE, Adjustment R^2, and (CC) values of the model are 3.79°, 0.96, and 0.98, respectively, which are better than traditional deep learning models such as Informer (4.14, 0.95, 0.98), CNN (5.56, 0.89, 0.96) and CNN-BiLSTM (3.97, 0.95, 0.98). In addition, the prediction time of our proposed model is only 11.67 $\pm$ 0.67 ms, which is less than 100 ms. Therefore, the real-time and accuracy of the model can meet the continuous prediction of human knee joint angle in practice.**

***Index Terms*—End-to-End, prediction, surface electromyography (sEMG), tightly coupled convolutional transformer(TCCT).**

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Ethics Committee of Xi'an Jiaotong University School of Medicine under Application No. 202206.

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I. Introduction

With the gradual application of wearable exoskeleton robots in human walking assistance and rehabilitation, the quality of life, as well as the motor ability of the users, have significantly been improved [1]. During exercise, the body usually selects specific muscles to control and adjust the angle of the joints to achieve the intended movement goal. The required muscles to control and adjust the joint angles can vary when completing different movements. Therefore, predicting knee joint angle can predict lower extremity movement intention. The traditional wearable exoskeleton robots have poor human-machine interaction and equipment synergy and cannot accurately predict human movement intentions in real-time. When the motion trajectory of the exoskeleton robot is inconsistent with that of the human body, it can hinder human motion and even cause secondary injury. Therefore, accurate and real-time recognition of human movement intention is the key to achieving perfect human-machine interaction and wearing the comfort of exoskeleton robots [2], [3], [4], [5].

At present, the human motion intention is mainly through the data obtained from Angle Sensors, Electroencephalogram (EEG)signals, or sEMG signals for prediction [6], [7], [8]. The data obtained by sensors are poor in real-time, which can not realize the regular and natural human-computer interaction. Since the motor encephalic region related to lower limbs is located deep in the brain, the collected scalp EEG has a weak amplitude. It is easily affected by the environment and users. Therefore, EEG is unsuitable as a signal source to identify the motor intention of lower limbs. sEMG is a bioelectric signal generated by the muscles' contraction on the human body's surface. It is directly related to the movement of the limbs and can be collected from the superficial muscles through electrodes [9]. It has the advantages of high resolution, mature acquisition technology, and non-invasive acquisition [10], [11], [12]. Therefore, recognizing human motion intention based on sEMG

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has gradually become one of the research hotspots in the field of the exoskeleton.

In past decades, researchers have mainly carried out sEMG-based human motion intention recognition by musculoskeletal models, traditional machine learning models, and deep learning models, as article [13], [14], [15]. However, the skeletal muscle model has many physiological parameters and poor prediction effects, which makes it difficult to apply in practice. Traditional machine learning methods require fewer features than skeletal muscle models to achieve better prediction results [16]. Still, most motion estimation methods based on traditional machine learning have complex feature extraction processes, poor robustness, and large prediction errors. Compared with conventional machine learning, deep learning-based motion intention recognition has high accuracy and better smoothness [17], [18]. However, the research related to deep learning focuses on the upper limb, while the research on the lower limb is little [19], [20], [21]. The most fundamental reason is that the lower limbs carry most of the human body's mass, which requires high real-time and accuracy.

Recently, much research about motion intention recognition focused on discrete motion classification and continuous motion regression based on deep learning [22], [23], [24]. A deep learning-based approach to discrete motion classification achieves good prediction results. However, the motion description of discrete motion classification is relatively simple, and there is no uniform classification standard [15], [25], [26], and it is impossible to predict actions outside the predefined. Therefore, discrete motion classification cannot be applied to the smooth control of wearable robots. Deep-learning-based continuous motion regression fuses traditional features of sEMG signals and elements from other multiple sources of information into the wearable robot control system to enable accurate angle prediction [27], [28], [29], [30]. However, too much manual intervention in this process and the long computation time of time-frequency domain features lead to low accuracy and real-time prediction [31], [32], [33].

To address the operational complexity of manual extraction of sEMG signal features and the limited prediction time. In this study, we propose an end-to-end method to predict human knee joint angles based on sEMG signals using a tightly coupled convolutional transformer (TCCT) model. First, we extracted the sEMG signal's envelope from five healthy subjects. There was no need to extract many features, simplifying the prediction process while reducing the interference of manually extracting features. After that, The sEMG signal envelope is then input into the TCCT model and trained by establishing a mapping between knee angle and sEMG signal envelope for continuous prediction of knee joint angle. The TCCT model combines the advantages of Informer and CNN to learn the deeper properties of signals. The results show that our end-to-end model can achieve higher accurate prediction and better real-time performance.

II. Method

A. Data Acquisition

The dataset used in this study was obtained from five non-disabled individuals. The inclusion criteria for the first four subjects were healthy adults between the ages of 18 and 26, who could walk independently, had no history of lower extremity pain or lower extremity injury in the past year, and had a completely normal neurophysiological examination. To illustrate the better predictive effect of our model on abnormal gait characteristics, we included subject 005 who had abnormal electrophysiological examination of the right leg and met the above criteria. He had slight foot drop in his right leg due to quadriceps weakness. The subject information table is listed in Table I. The Ethics Committee of Xi'an Jiaotong University School of Medicine approved the experiment (202206). All procedures were performed by the ethical standards of the 1964 Declaration of Helsinki and its later amendments. All subjects gave written informed consent and agreed to use their anonymous data for scientific and educational purposes. Participation was voluntary and could be terminated at any time.

TABLE I
Subjects' Basic Information Table

Subject No.	Gender	Height(cm)	Weight(kg)	Age
001	Male	172	60	23
002	Male	167	75	26
003	Female	160	41	25
004	Female	161	50	24
005	Male	166	60	23

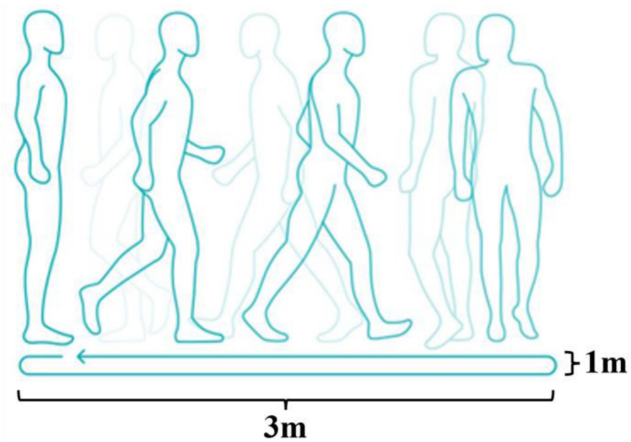

Fig. 1. Walking route of the "two-minute walk experiment".

All the experiments were conducted at the Gait Analysis Laboratory at the Gait Lab, Xi'an Jiaotong University Hospital. Based on the actual human walking scene, we designed a "two-minute walking experiment" to collect sEMG data during normal human walking and turning. Fig. 1 shows the walking route. Each subject was asked to walk barefoot at their natural speed. All walking trials were conducted under the guidance of an experienced physiotherapist. The subjects were fully rested between different trials to prevent muscle fatigue. Each subject completed at least 6 successful gait experiments. Five channels of sEMG signals were collected at 2000 Hz by Noraxon Ultium EMG wireless surface electromyography (Noraxon, USA). The unilateral sEMG acquisition sites are shown in Fig. 2. The monitored muscles involved the lateral femoral, soleus, tibialis anterior, medial gastrocnemius, and biceps femoris of both lower limbs. The sweat hairs at the location of the pasted electrodes

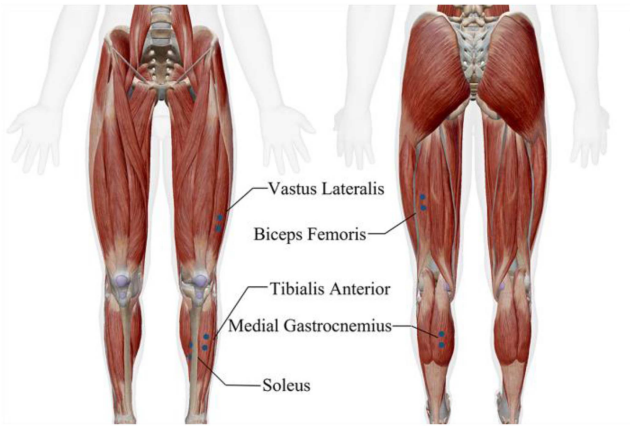

Fig. 2. Left lower limb sEMG acquisition site. The right side uses the corresponding acquisition position, both using double electrodes to collect the surface area electrical signal to reduce external interference during the measurement.

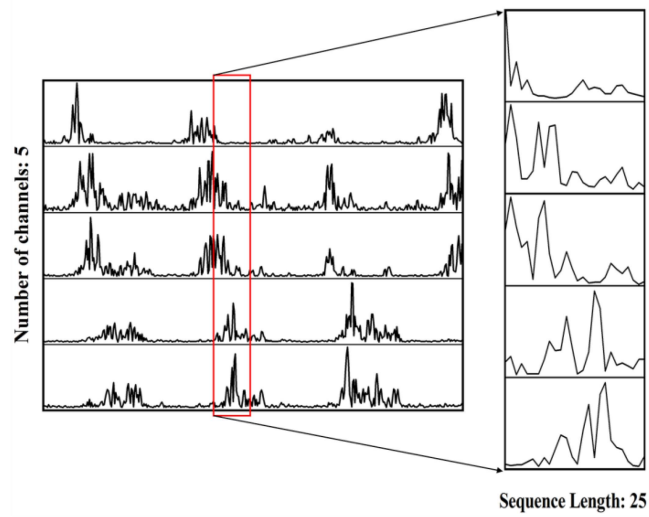

Fig. 3. Schematic diagram of the model input signal.

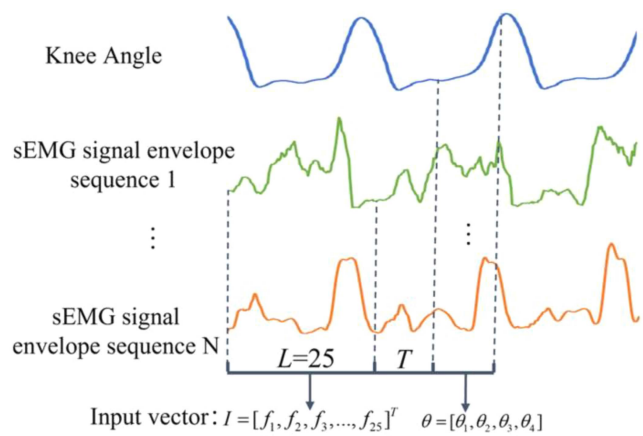

Fig. 4. Prediction data calibration schematic. The curves in the figure represent the knee angle and the 5-channel sEMG signal envelope, where L denotes the length of the input time series, and T (100 ms) denotes the prediction time.

were scraped off, and the dead skin was rubbed off with a unique scrub, followed by an alcohol wipe to reduce noise interference and obtain stable quality sEMG. The lower limb joint resolution system obtained the joint angle signals. The system was based on the Qualisys 3D motion acquisition and analysis system (Qualisys, Switzerland). The joint angles of the human lower limbs were solved in Visual 3D software using reflective marker points attached to the human body and the QTM data management software.

B. Data Pre-Processing

In the sEMG envelope extraction process, we removed the noise using a 4th-order Butterworth bandpass filter at 20–350 Hz, followed by a 4th-order Butterworth trap filter at 50 Hz to remove the industrial frequency interference. Since the sampling frequency of sEMG and joint angle signals were inconsistent, we resampled them all to 100 Hz. Finally, we performed full-wave rectification and fourth-order Butterworth low-pass filtering (fc = 10 Hz) on the sEMG to extract the signal envelope.

In this study, the input signals were combined into a matrix with the row vectors representing the linear envelope values of the sEMG signals from the five channels and the column vectors representing the time series of the sEMG signal envelope with a sequence length of 25. We took 25 consecutive sEMG signals linear envelope values as model input to obtain sufficient data and a suitable prediction interval. The schematic diagram of the model input signal is shown in Fig. 3. The sEMG signals channels we chose correspond to the most superficial muscle and feature combinations. Each input signal is represented by a matrix I of size 25×5: $I = [f_1, f_2, f_3, \ldots, f_{25}]^T$, $f_i = [PF_{TA,i}, PF_{MG,i}, PF_{SOL,i}, PF_{BF,i} PF_{VL,i}]^T$, where f_i denotes the 5-channel sEMG signals envelope signal extracted at the ith moment, and PF indicates the sEMG signals linear envelope value. The beginning moments of every two inputs time series were separated by a step length (10 ms). The input signals are calibrated as in Fig. 4. Each set of input signals corresponds to the label of the lower limb knee angle $\theta = [\theta_1, \theta_2, \theta_3, \theta_4]$ at four consecutive time points after the prediction time.

Z-score normalization was applied to each channel of the recorded signals to keep all eigenvalues under the same magnitude and speed up the model convergence. The processed data conformed to the standard normal distribution, the mean value was 0, and the standard deviation was 1. The transformation function is shown in (1).

$$Z = \frac{x - \mu}{\sigma} \tag{1}$$

where Z is the standardized value, x is the observed value of the sample, μ is the mean of data, and σ is the standard deviation of data.

C. TCCT Model Structure

The framework of sEMG signal training and prediction by the TCCT model is shown in Fig. 5. The processed sEMG signal

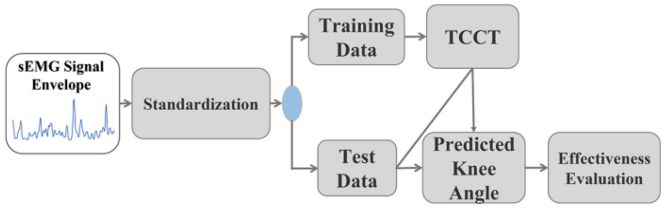

Fig. 5. Framework for training and prediction of sEMG signal by TCCT model.

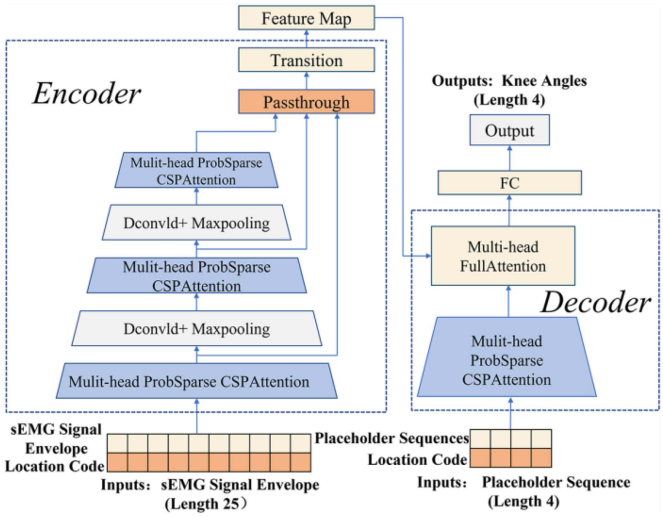

Fig. 6. TCCT knee joint angle prediction model.

envelope is normalized to obtain the dataset for the model input, divided into two parts: the training set and the test set. The training set is applied to train the TCCT model, and the test set is used to test the model training effect.

1) TCCT Model: Knee angle prediction using sEMG signals is similar to the time series prediction problem. However, the input and output sequences for the time series prediction problem are the same variable ordinarily. In contrast, the input and output sequences for the knee angle prediction problem are not the same variable, which makes our task relatively difficult. In this study, we mainly used the TCCT model [34] proposed by Shen et al. to achieve knee joint angle prediction. It combines Informer, CSPAttention, Dilated Causal Convolution, and Passthrough mechanism. The TCCT knee joint angle prediction model structure is shown in Fig. 6. The TCCT model mainly comprises two parts: encoder and decoder. The encoder part stacks three Mulit-head ProbSparse CSPAttention layers connected by Dilated Causal Convolution and Maxpooling layers. The feature sequences of different lengths obtained by the Mulit-head ProbSparse CSPAttention layer are stitched together by the Passthrough mechanism to obtain the final feature map. The decoder consists of Mulit-head ProbSparse CSPAttention and Multi-head FullAttention. Since the input and output are not the same variables, the model decoder input has only the placeholder sequence with all elements as 0. Meanwhile, the masked self-attention in the first layer of the decoder is changed to the standard self-attention.

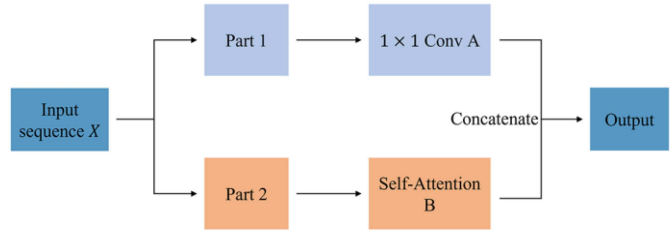

Fig. 7. CSPAttention block. The input sequence X is divided into two parts: the first part (blue) passes through a 1 × 1 convolution layer, and the second part (red) passes through a standard self-attentive layer. The two parts of the output are joined together as the output of CSPAttenation.

2) Informer Module: The Informer is a derivative of the standard Transformer model. Compared to the standard Transformer model, the Informer model has been improved in the following ways. 1) The self-attention mechanism of the specification in the encoder is replaced by the ProbSparse self-attention mechanism, which improves the query efficiency. 2) Connecting self-attention blocks using convolutional and max-pooling layers allows stacking more self-attention blocks. 3) The ability to parallelize k-1 encoders enhances robustness. 4) The masked ProbSparse self-attention mechanism replaces the masked self-attention mechanism of the original decoder, and the decoder input is composed of two parts: the start token and the target sequence. These changes enable Informer to perform optimally in the Transformer-like model. For details related to Informer, please refer to Zhou et al. [35].

3) CSPAttention: The CSPAttention block is shown in Fig. 7. CSPAttention is similar to the structure of CSPNet [36]. The input sequence $X \in R^{L\times d}$ is divided into two parts $X = [X_1^{L\times d_1}, X_2^{L\times d_2}]$ in the dimensional direction, where L is the input length and d is the input dimension. X1 is connected to the end of the module after passing through a 1 × 1 convolutional layer A, while X2 acts as the input of the self-attention module B. The outputs of A and B are connected by dimensional transformation and are used as the output of the whole module. Using CSPAttention can alleviate the self-attention mechanism's memory bottleneck and computational efficiency problems.

4) Dilated Causal Convolution: Since the standard convolutional layer in Informer cannot provide a large enough perceptual field, the Dilated Causal Convolution is used in TCCT to increase the perceptual field instead of the standard convolutional layer. The Dilated Causal Convolution operation C (with kernel size k) for the elements $x_n \in R^d, n \in L$ in the sequence $X \in R^{L\times d}$ on the ith convolution layer is defined as

$$C(x_n) = \begin{bmatrix} x_n \\ x_{n-i} \\ \cdot \\ \cdot \\ \cdot \\ x_{n-(k-1)\times i} \end{bmatrix} W^{d\times d'} \tag{2}$$

where d' is the output dimension; i is the expansion factor.

The two adjacent convolution kernels of the ith layer of the Dilated Causal Convolution layer are separated by $(2^{i-1} - 1)$

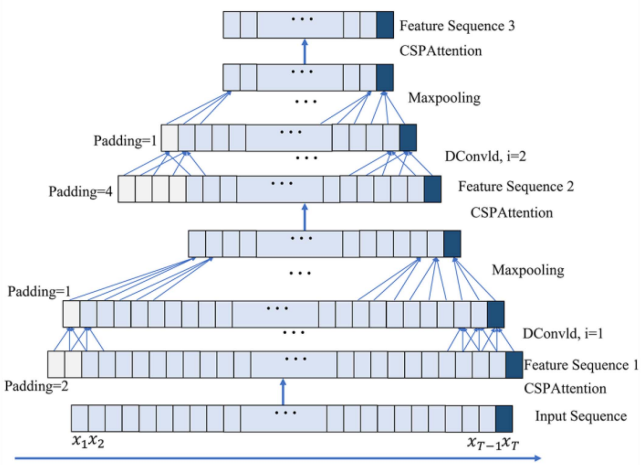

Fig. 8. Visualization graph of the combination of Dilated Causal Convolution and CSPAttention. Three CSPAttentions are stacked in the figure and connected by Dilated Causal Convolution and Maxpooling layer. The figure stacks 3 CSPAttentions, which are connected by a Dilated Causal Convolution and a Maxpooling layer.

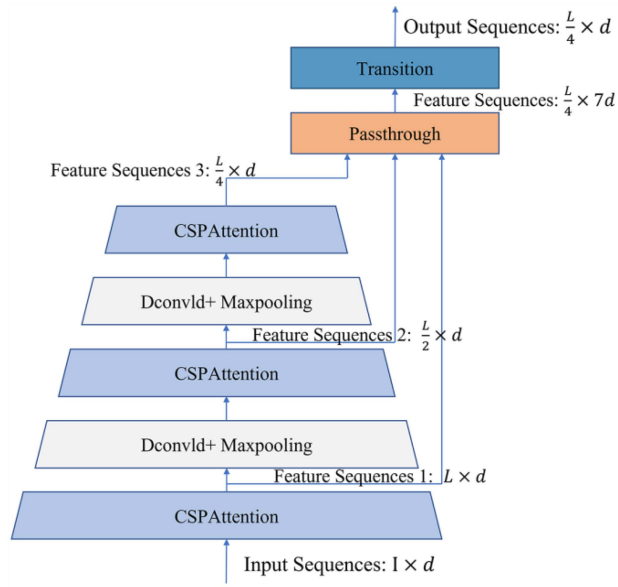

Fig. 9. Passthrough mechanism visualization of connecting feature sequences of different lengths. The figure stacks 3 CSPAttention layers connected by Dilated Causal Convolution and Maxpooling layers. The Passthrough mechanism stitches together feature sequences of different lengths.

elements. Each element X in the sequence at time t is only convolved with the element at or before time t to ensure no information is missed when the convolution operation is performed. When i = 1, the Dilated Causal Convolution is the same as the standard causal convolution. Fig. 8 shows the visualization schematic of the three CSPAttentions blocks stacked with the Dilated Causal Convolution layer (convolution kernel size of 3).

5) Passthrough Mechanism: The Yolo series of target detection CNN [36], [37], [38] networks propose a Passthrough mechanism similar to the feature pyramids in computer vision and CNN [39], applying the Passthrough mechanism with Informer to merge feature sequences of different scales to obtain more comprehensive information. An encoder with CSPAttendant and Dilated Causal Convolution applied stacked with n self-attentive modules will produce n feature sequences, the kth $(k = 1, 2 \ldots, n)$ feature sequences has length $L/2^{k-1}$ and dimension d. The kth feature sequence is divided into 2^{n-k} feature sequences of length 2^{n-k} by the Passthrough mechanism and all equal-length sequences are concatenated. As shown in Fig. 9, the Passthrough mechanism visualization of connecting feature sequences of different lengths. This way, the effect of multiple encoders in parallel can be achieved using an Informer with one encoder.

D. Evaluation

In the experiment, each subject's first 10 minutes of gait data were used as the training and validation sets (training set: validation set = 9:1). The remaining 2 minutes of gait data were used as a separate test set. To further evaluate the effectiveness of our model, we used RMSE, CC, and Adjustment R^2 as metrics to assess the error between the actual knee angle and the predicted knee angle. RMSE represents the error between the actual and predicted values. CC represents the similarity between the precise knee angle and the expected knee angle shape, and the CC value ranges from [0, 1]. The closer the result is to 1, the higher the correlation between the two variables. We chose Adjustment R^2 to eliminate the effect of sample size. They are calculated as follows.

$$RMSE = \sqrt{\frac{1}{N}\sum\left(\theta_{pre} - \theta_{real}\right)^2} \tag{3}$$

$$CC = \frac{\mathrm{cov}(\theta_{pre}, \theta_{real})}{\sigma_{pre}\sigma_{real}} \tag{4}$$

$$R^2 = 1 - \frac{\sum\left(\theta_{pre} - \theta_{real}\right)^2}{\sum\left(\theta_{real} - mean\left(\theta_{real}\right)\right)} \tag{5}$$

$$\text{Adjustment } R^2 = 1 - \frac{\left(1 - R^2\right)(N-1)}{N-p-1} \tag{6}$$

where θ_{pre} is the predicted knee angle; θ_{real} is the real knee angle; N is the number of samples. $\mathrm{cov}(\theta_{pre}, \theta_{real})$ is the covariance between the actual angle and the predicted angle; σ_{pre} is the standard deviation of the true angle; θ_{real} is the standard deviation of the predicted angle; N is the number of samples; p is the number of features.

E. Experimental Details

The experiment was conducted with the Pytorch deep learning framework in Python. Epoch = 20, batch size = 32, and the optimizer was chosen as Adam. The initial learning rate of the training phase is 0.0001. We chose MSE between the predicted and actual angles as loss functions. In addition, the encoder input is a 5-channel sEMG signals envelope sequence of length 25, the decoder input was a placeholder sequence with all elements 0 of length 4, and the decoder output was the knee angle after 100 ms of length 4. All experiments were performed on an NVIDIA GeForce RTX 2080 Ti GPU.

TABLE II
KNEE JOINT ANGLE PREDICTION RESULTS OF DIFFERENT MODELS

Model	RMSE (°)	Adjustment R^2	CC
Informer	4.14	0.95	0.98
TCCT1	4.05	0.95	0.98
TCCT2	3.99	0.96	0.98
TCCT	**3.79**	**0.96**	**0.98**

The TCCT model gives the best results compared to CNN, BiLSTM, and CNN-BiLSTM and puts its RMSE, Adjustment R2, and (CC) values in bold.

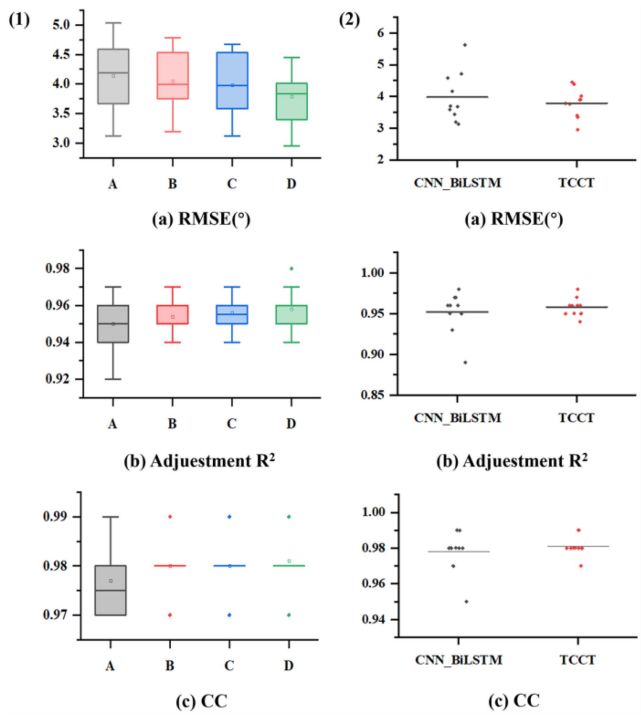

Fig. 10. (1) Comparison of prediction results of different models. (a) RMSE, (b) Adjustment R^2, (c) CC, where A denotes Informer, B denotes TCCT1, C denotes TCCT2, and D denotes TCCT. (2) Scatter plot of CNN-BiLSTM and TCCT prediction results. (a) RMSE, (b) Adjustment R^2, (c) CC.

III. RESULTS

A. Analysis of TCCT Model Results

To compare the effects of different model blocks on joint angle prediction, we designed three additional models like Shen et al. [34]. Informer, the most basic Informer model with only one encoder; TCCT1, Informer model combined with CSPAttention; TCCT2, Informer model combined with CSPAttention and dilated casual convolution. Model evaluation was performed using RMSE, Adjustment R^2, and CC. Table II shows that the mean values of RMSE, Adjustment R^2, and CC of the TCCT model for all five healthy subjects were 3.79, 0.98, and 0.96, and our model achieved better prediction than other models. Fig. 10(1)(a)–(c) represent the RMSE, Adjustment R^2, and CC of different models, respectively. Among the prediction results of all models, the Adjustment R^2 and CC of all models except CNN are similar to those of TCCT, but the RMSE of TCCT results are significantly lower than those of other models.

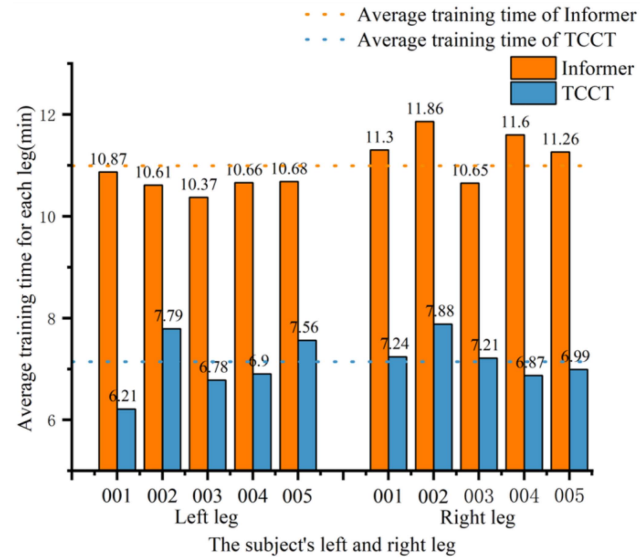

Fig. 11. Average time consumed by different models trained 10 times. The data for each leg of all subjects in the figure were trained 10 times in both models.

To compare the computational efficiency of the Informer and TCCT models, we trained the data from all subjects for each leg 10 times using the Informer and TCCT models, and the measured average training time for each leg using the Informer and TCCT models is shown in Fig. 11. In addition, a matched-samples t-test of the mean measurement time was used to test whether the CSPAattention structure could significantly reduce the training time. The data show that the training time of the Informer model ($M = 10.99$, $SD = 0.49$) is higher than that of the TCCT model ($M = 7.14$, $SD = 0.51$), and this difference is statistically significant, $t(9) = 20.33$, $p < 0.001$, $d = 6.4$, 95%CI [3.41;4.27]. Combined with Table II, we can see that the model with CSPAttention can improve the computational efficiency with the same data input, and the use of Dilated Causal Convolution and Passthrough mechanism not increase the additional arithmetic power consumption but also improves the prediction results.

In addition, we randomly selected a portion of the data from each leg of subjects 001 to 005 as the test set to visualize our model's performance. Fig. 12 shows the visual comparison of the actual knee angle and the predicted knee angle for subjects 001 to 005. We can observe that the prediction results of our proposed TCCT model are very close to the real angle. It is worth noting that the shape of the knee angle curve of subject 005's right leg in the above figure is different from the others, which does not match the trend of knee angle change in average humans walking on flat ground. After our study, we found that our method also has good prediction for the data of 005 subject who have abnormal gait. Therefore, our proposed TCCT model has a great application value for predicting the knee joint angle of human lower limbs.

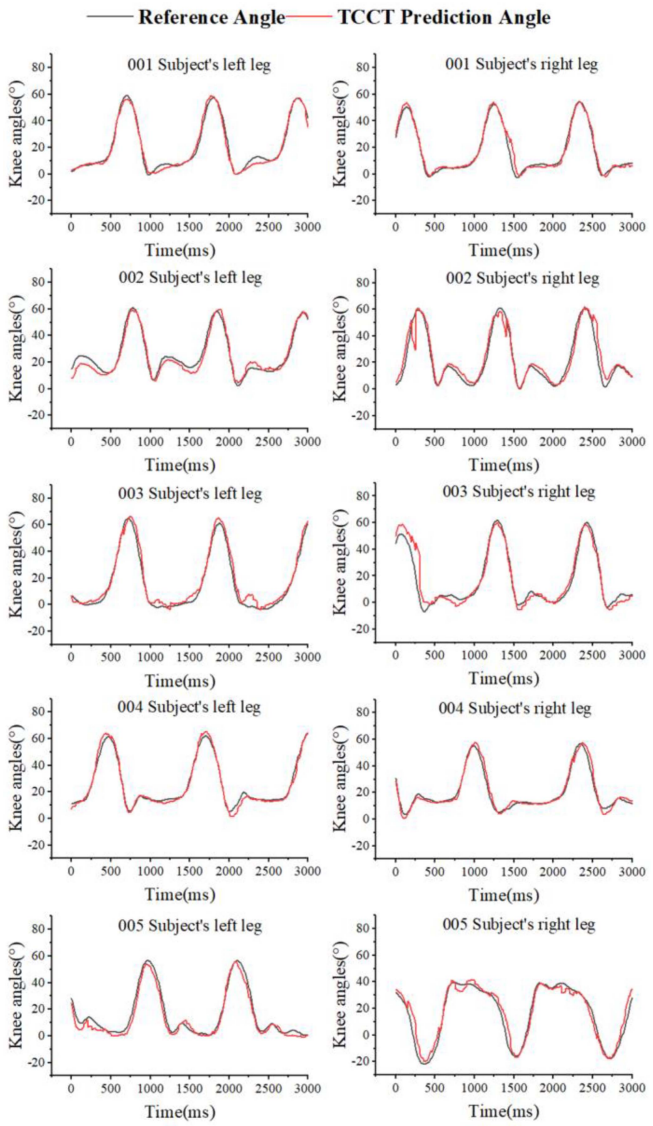

Fig. 12. Visualization of human knee joint angle prediction curve based on the TCCT model. The subplots in the same row correspond to the subject's left and right leg knee angle curves, and the subplots in different rows correspond to the knee angle curves of different subjects.

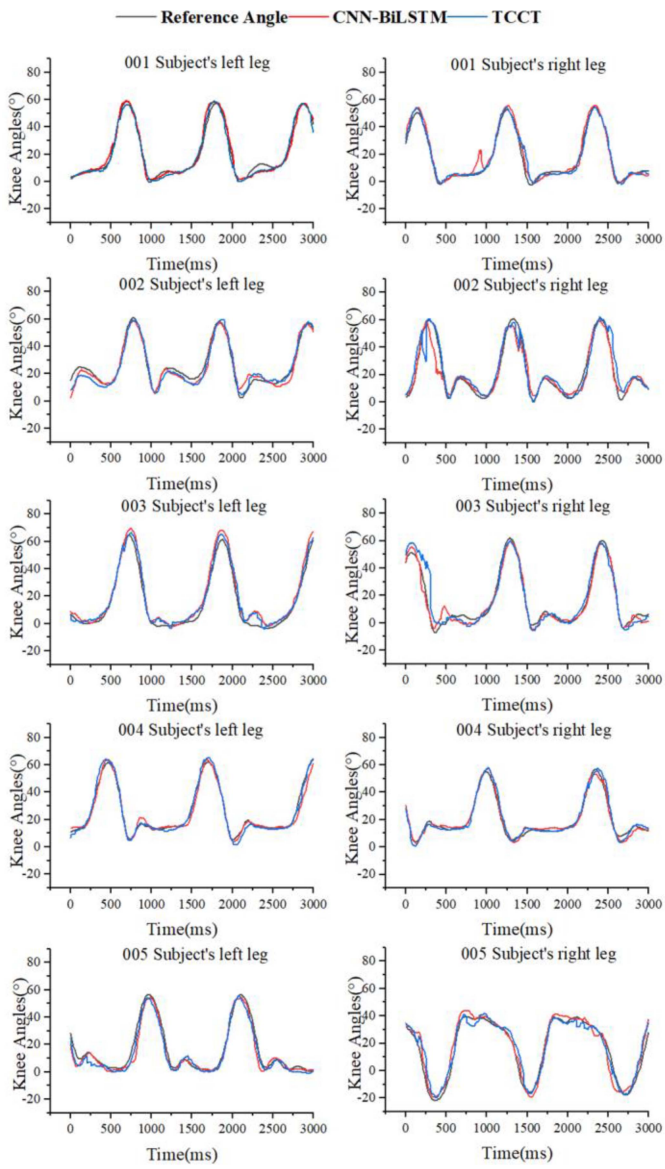

Fig. 13. Visual comparison of prediction results of CNN-BiLSTM and TCCT models.

TABLE III
KNEE JOINT ANGLE PREDICTION RESULTS OF DIFFERENT MODELS

Model	RMSE (°)	Adjustment R^2	CC
CNN	5.56	0.89	0.96
BiLSTM	4.09	0.94	0.97
CNN-BiLSTM	3.97	0.95	0.98
TCCT	**3.79**	**0.96**	**0.98**

The TCCT model gives the best results compared to CNN, BiLSTM, and CNN-BiLSTM and puts its RMSE, Adjustment R2, and (CC) values in bold.

B. Comparison With Previous Work

To further illustrate the advantages of our proposed model, we compare previous work with the TCCT model, where we input the sEMG signal envelope into CNN, BiLSTM and CNN-BiLSTM models [41]. Table III shows the RMSE, Adjustment R^2, and CC average in different models. As can be seen from Table III, the TCCT model still has the best prediction performance compared with other methods. We used the CNN-BiLSTM model as a baseline to visualize and compare the motion intention with the TCCT model, and the comparison results are shown in Fig. 13. The figure shows that the TCCT model achieves similar results to the CNN-BiLSTM model, with some subjects having smoother predictions closer to the reference angle. Fig. 10(2) shows the scatter plots of the prediction results of the CNN-BiLSTM model and the TCCT model, with Fig. 10(2)(a)–(c) indicating the RMSE, Adjustment R^2, and CC, respectively. The figure shows that compared with the CNN-BiLSTM model, TCCT is better in all evaluation metrics, and the scatter distribution is more concentrated with fewer outliers, indicating that the TCCT model's prediction results are more accurate and stable.

C. Real-Time Validation of TCCT Model Predictions

Human lower limb continuous motion prediction requires higher real-time performance. If the real-time performance of the algorithm is ignored and the improvement of prediction accuracy is pursued, it will instead cause a hindrance to human walking. In this experiment, to initially test the real-time performance of the algorithm, a Python program was designed to integrate all the steps of signal reading, signal pre-processing, and outputting test results based on the trained TCCT model to calculate the prediction time. The results of ten experiments showed that the average training time of the TCCT model was 7.14 ± 0.51 minutes, and the prediction time was 11.67 ± 0.67 ms. This test time is less than the prediction time of the CNN-BiLSTM model of 62.47 ± 0.29 ms and the fastest time of 30 ms generated by sEMG [12]. The results show that our model can predict the knee joint angle before human motion and leave time for other mechanical delays. Therefore, the real-time performance of the experiment meets the requirements.

IV. Discussion

In this study, we propose to use the TCCT model for end-to-end prediction of human knee angle, which reduces the human interference caused by manual extraction of sEMG signals, improves the prediction accuracy, and shortens the prediction time. The results showed that the RMSE of our method was 3.79, and the prediction time was only 11.67 ± 0.67 ms. Notably, From Fig. 11 we can see that the gait characteristics of the right leg of subject 005 are significantly different compared to normal. Still, the same good prediction results were obtained, indicating that this model could predict the knee joint angle for abnormal gait. The results in Fig. 12 show that it can improve the wearing comfort to enhance the human-computer interaction experience [2], [3], [4], [5].

We used the sEMG signal envelope obtained by simple processing as input to achieve end-to-end prediction of knee joint angle. From the results in Table II, the TCCT that combines the three modules has the best prediction performance. Although the Adjustment R^2 and CC are similar to the TCCT1 and TCCT2, the RMSE of the TCCT is significantly lower than the other models. The prediction performance of all three TCCT models is better than the Informer model, indicating that the three TCCT structural modules improved Informer's time series prediction ability. Combining the results of Fig. 11 average training time, TCCT, TCCT1, and TCCT2 have better prediction and shorter training time than Informer, indicating that CSPAttention can mitigate the Informer structure without losing accuracy and Dilated Causal Convolution and Passthrough can further improve the prediction ability and reduce the prediction error [1]. This is consistent with the findings of Shen et al. [34].

Compared with CNN, BiLSTM, and CNN-BiLSTM, the TCCT model has slightly better prediction results and more concentrated result distribution, indicating that its prediction results are more stable [41]. This phenomenon is because CNN can only learn local spatial features of the input vector, and BiLSTM has no long-time dependence to capture the periodicity of time series. Compared to CNN or RNN time series prediction models, Informer's self-attention mechanism is equally available for any input component independent of temporal distance, which makes Informer models more effective in handling long-range information and capturing long-range dependencies. Moreover, TCCT combines CNN and Informer model, entirely using the complementary characteristics of CNN local feature extraction and Informer capturing long-range information to improve the learning capability further.

sEMG will be generated 30–150 ms before the action [12], we use this theory to predict knee joint angle in human lower limbs. Our model takes only 11.67 ms to predict the knee joint angle at the next moment, which fully satisfies the requirement of real-time and leaves enough time for other mechanical delays. Therefore, the prediction model designed in this study meets the real-time requirement in practical applications. We speculate this is because Transformer is not a sequential structure, and its self-attention mechanism is not affected by the time distance, so it has better parallelism and requires less prediction time.

The above experimental results demonstrate the validity of the end-to-end knee angle prediction model TCCT, but the method also has some limitations. First, this study only evaluates the model in the offline case for prediction accuracy and prediction time without actually applying the model to the control system of an exoskeleton robot and lacks further validation, which will be carried out subsequently. Second, all of our data sources came from normal individuals around 24 years old. Although one subject had abnormal gait characteristics in the right leg, the experimental data sources lacked diversity. The follow-up study focused on validation and optimization for the gait abnormality problem. Finally, although the end-to-end TCCT prediction model has good prediction results [23], [42], [43], it is a black-box model with low explanatory power, which is not conducive to the algorithm's generalization. In the future, we will further design white box models with strong explanations and apply them to online tests for the rehabilitation training of patients with lower limb dysfunction.

V. Conclusion

Accuracy and real-time play an essential role in predicting human lower limb movement intention and improving the coordination between exoskeleton robots and human lower limbs. In this study, an end-to-end knee angle prediction model using TCCT is proposed, which uses a simple processed sEMG signal envelope as input to improve the accuracy of knee angle prediction while reducing manual intervention. The results showed that the average of the model's RMSE, Adjustment R^2, and CC were 3.79°, 0.96, and 0.98, respectively, with a prediction time of 11.67 ± 0.67 ms. Our model has more minor errors and shorter prediction times than established models based on the same topic and allows more time for other machine delays. In comparison with TCCT1 and TCCT2, we found that each module of TCCT is essential, and the lack of any one module will affect the accuracy of the prediction. Although the model is relatively complex and requires a long training time, it has the advantages of a good forecast, a short prediction time, and no manual feature extraction. This proves that the model has great potential to improve the human-computer interaction of rehabilitation robots.

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